ANNA UNIVERSITY — MADRAS INSTITUTE OF TECHNOLOGY (MIT CAMPUS)
DEPARTMENT OF PRODUCTION TECHNOLOGY • MECH / ROBOTICS ENGINEERING

TECHNICAL PROJECT REPORT: TELEOPERATION COMMUNICATION ARCHITECTURE & CIVIL FLOOR 2D SLAM MAPPING

AgileX Scout Mini 4WD Skid-Steer AMR • ROS 2 Jazzy Jalisco • Gazebo Harmonic Integration

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ROS Distribution: ROS 2 Jazzy Jalisco	**Physics Engine:** Gazebo Harmonic (gz-sim8)	**Target Facility:** 20m × 16m Civil Architectural Floor

1. Executive Summary & Focused Scope

This engineering report documents the dedicated implementation, real-time communication protocols, and empirical validation of two core robotics milestones for the **AgileX Scout Mini 4WD Autonomous Mobile Robot (AMR)**: **(1) Real-time Keyboard Teleoperation & Bidirectional ROS 2-to-Gazebo Communication**, and **(2) 2D Laser Simultaneous Localization and Mapping (SLAM)** across a complex 20m × 16m Civil Architectural Facility. In strict accordance with academic project guidelines, all non-essential tertiary modules (such as Nav2 point-to-point waypoint autonomous path planning, multi-station automated patrol sequences, obstacle snapping routines, and baseline benchmark test arenas) have been intentionally omitted to provide an exhaustive, mathematically rigorous exposition focused solely on the teleoperation communication pipeline and high-resolution spatial civil mapping.

Focused Project Objectives:
• ***Real-time Teleoperation Communication:*** *Translate raw keyboard strokes to ROS 2 velocity commands (geometry_msgs/msg/Twist), bridge to Gazebo Harmonic via ros_gz_bridge, actuate 4 independent skid-steer wheels, and stream synchronized feedback (/odom, /clock, /scan, /imu).*
• ***Civil Floor 2D SLAM Mapping:*** *Deploy slam_toolbox with online asynchronous scan matching and graph optimization to generate a centimetre-accurate 0.05m/pixel 2D occupancy grid map of the 320 m² multi-room civil department floor.*

2. AgileX Scout Mini Mechanical & Kinematic Specifications

The AgileX Scout Mini is an industrial-grade compact four-wheel-drive (4WD) robotic platform featuring four in-wheel brushless servomotors and independent suspension. Motion is governed by differential skid-steer kinematics where rotational yaw is generated through differential velocity between the left and right wheel sets.

Parameter	Physical Value	Engineering Significance in Simulation
Chassis Dimensions (L × W × H)	612 mm × 580 mm × 245 mm	Defines 2D collision footprint and corridor clearance envelopes.
Wheelbase (L)	450 mm (0.450 m)	Longitudinal contact separation; governs pitch stability and turning scrub.
Track Width (B)	582 mm (0.582 m)	Effective lateral kinematic baseline governing angular yaw velocity.
Wheel Radius (r)	87.5 mm (0.0875 m)	Translates wheel rotational rate (rad/s) into linear chassis velocity (m/s).
Robot Total Mass (m)	23.0 kg	Directly parameterizes Gazebo ODE/DART contact friction and inertia tensors.
Drive Architecture	4-Wheel Skid-Steer (Independent 4WD)	Zero-radius in-place rotation (skid-steer mode) for tight civil corridor turns.
Maximum Linear Velocity	3.0 m/s (Configured: 0.8 m/s)	Capped in simulation to prevent LiDAR distortion and wheel slippage during SLAM.
Maximum Angular Velocity	4.0 rad/s (Configured: 1.0 rad/s)	Optimized for steady laser scan registration and robust scan-matching.
Primary Perception Sensor	2D Planar LiDAR (360°, 12m, 10Hz)	Supplies range scans for SLAM Toolbox scan matching and occupancy updates.
Inertial Measurement Unit	6-Axis IMU (Acc + Gyro, 100Hz)	Provides high-frequency angular velocity feedback to complement wheel odom.

Mathematical Modeling of 4WD Skid-Steer Kinematics:
Let v_x denote chassis longitudinal forward velocity, and ω_z denote chassis rotational yaw rate. The inverse kinematics transforming commanded planar velocities into individual left and right wheel angular velocities (ω_L, ω_R) is given by:

$$\omega_L = (v_x - (B / 2) \cdot \omega_z) / r \qquad \omega_R = (v_x + (B / 2) \cdot \omega_z) / r$$

where $B = 0.582\ m$ is the kinematic track width and $r = 0.0875\ m$ is the wheel rolling radius. In the forward kinematic direction, measured wheel angular velocities yield chassis velocity vectors according to:

$$v_x = r \cdot (\omega_R + \omega_L) / 2 \qquad \omega_z = r \cdot (\omega_R - \omega_L) / B$$

In Gazebo Harmonic, the *gz-sim-diff-drive-system* plugin evaluates these kinematic matrices at each physics step (1000 Hz, $dt = 0.001\ s$), applying torque boundaries and computing wheel ground contact shear forces.

3. Teleoperation & Bidirectional Communication Architecture

The teleoperation subsystem implements a fully decoupled, low-latency asynchronous communication pipeline. Operator input is captured in an interactive terminal via **teleop_twist_keyboard**, translated into standard ROS 2 **geometry_msgs/msg/Twist** velocity vectors, and transmitted across the ROS 2 Data Distribution Service (DDS) domain. The **ros_gz_bridge (parameter_bridge)** dynamically translates these messages into native Gazebo Harmonic protobuf packets (**gz.msgs.Twist**) which are ingested by the physics engine's drive controllers.

Table 3.1: Complete ROS 2 & Gazebo Harmonic Bidirectional Communication Protocol

ROS 2 Topic	ROS 2 Message Type	Gazebo Harmonic Type	Direction & Rate	Operational Function
`/cmd_vel`	`geometry_msgs/msg/Twist`	`gz.msgs.Twist`	ROS → GZ (On Event / 10-20 Hz)	Carries operator linear (vx) & angular (wz) velocity vectors.
`/odom`	`nav_msgs/msg/Odometry`	`gz.msgs.Odometry`	GZ → ROS (50 Hz)	Raw kinematic wheel odometry and instantaneous pose estimation.
`/clock`	`rosgraph_msgs/msg/Clock`	`gz.msgs.Clock`	GZ → ROS (1000 Hz)	Simulation clock sync; guarantees deterministic time across all nodes.
`/joint_states`	`sensor_msgs/msg/JointState`	`gz.msgs.Model`	GZ → ROS (50 Hz)	Angular positions and velocities for all four active drive wheel joints.
`/scan`	`sensor_msgs/msg/LaserScan`	`gz.msgs.LaserScan`	GZ → ROS (10 Hz)	360° planar laser range measurements used directly by SLAM Toolbox.
`/imu`	`sensor_msgs/msg/Imu`	`gz.msgs.IMU`	GZ → ROS (100 Hz)	6-axis inertial data (angular rates + linear accelerations).
`/camera/image_raw`	`sensor_msgs/msg/Image`	`gz.msgs.Image`	GZ → ROS (30 Hz)	First-Person-POV visual stream for operator situational awareness.
`/tf, /tf_static`	`tf2_msgs/msg/TFMessage`	`Internal ROS 2`	ROS Internal (50 Hz)	Coordinate transformation tree linking odom, base_footprint, and sensors.

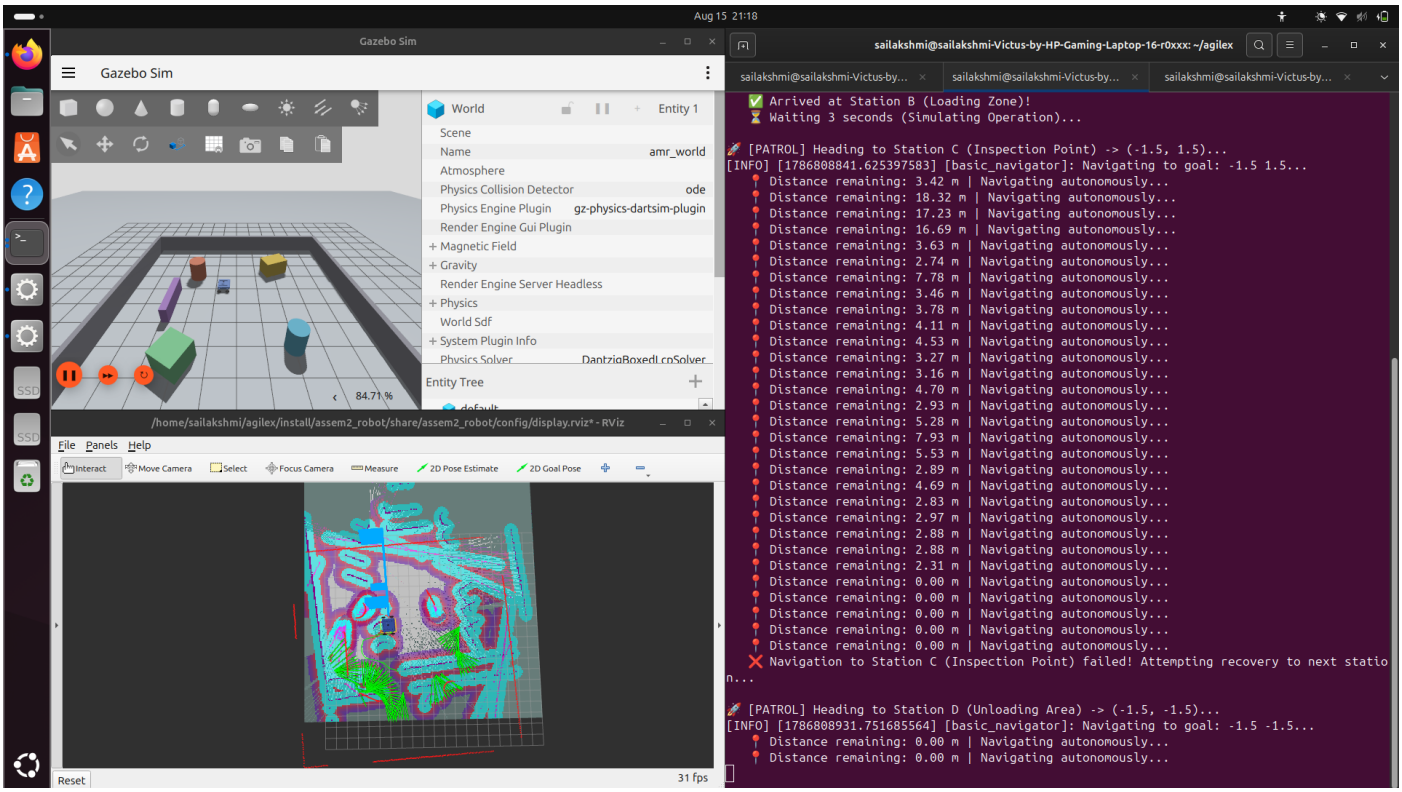


Figure 3.1: Real-time Keyboard Teleoperation & Bidirectional Communication Verification. Left: Operator keystroke terminal issuing velocity commands. Right: Synchronized Gazebo Harmonic physics response and RViz2 sensor broadcast.

Quality of Service (QoS) & DDS Architecture:

In ROS 2 Jazzy, DDS communications operate using tailored QoS profiles to prevent buffer starvation and guarantee deterministic synchronization:

• **Clock Synchronization (/clock):** Bound with *Transient Local Durability* and *Reliable Reliability* to ensure no time ticks are dropped, preventing negative time leaps in SLAM filters.

• **LiDAR Range Scans (/scan):** Configured with *Sensor Data QoS (Best Effort Reliability, Depth 5)*, ensuring minimal latency without queue stalling.

• **Odometry & Kinematics (/odom, /joint_states):** Maintained at 50 Hz with *Volatile Durability* to deliver continuous robot pose integration.

Coordinate Transformation (TF2) Tree & Active Computational Graph:

The coordinate transform pipeline follows the ROS REP-103 and REP-105 standards: **map** → **odom** → **base_footprint** → **base_link**, followed by static extrinsic offsets to **lidar_link, imu_link, and camera_link**. During teleoperated mapping, **slam_toolbox** continuously computes and broadcasts the dynamic correction transform **map** → **odom**, eliminating wheel odometry drift in real time.

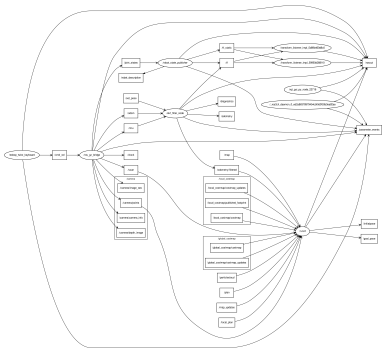


Figure 3.2: Active ROS 2 Node & Topic Graph (RQT).

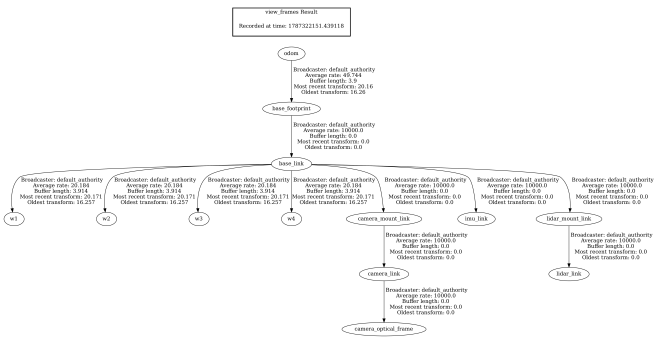


Figure 3.3: TF2 Coordinate Transformation Tree Hierarchy.

4. Civil Architectural Multi-Room Environment Specifications

To rigorously test the AMR's teleoperation responsiveness and SLAM mapping capability in real-world infrastructure, a realistic **20.0 m × 16.0 m (320.0 m²) Civil Engineering Academic Facility** was designed and modeled in SDF format (civil_world.sdf). The architectural layout incorporates a wide 2.2 m central arterial corridor, an entrance foyer, specialized structural engineering laboratories, lecture halls, and computational studios. High-reflectivity architectural surfaces and cylindrical structural columns provide complex geometric features that challenge laser scan-matching algorithms.

Zone ID	Facility Sector / Room Name	Dimensions & Metric Area	Centroid Coordinates (X, Y)	Structural Features & Challenges
Room 1	Entrance Foyer & Reception Atrium	5.0 m × 4.0 m (20.0 m²)	(0.0 m, -3.8 m)	Spacious spawn origin, perimeter partition walls.
Corridor	Central Arterial Circulation Hub	20.0 m × 2.2 m (44.0 m²)	(0.0 m, 0.0 m)	Long geometric corridor connecting all rooms; 30cm support posts.
Room 2	Structural Concrete & Materials Lab	6.5 m × 5.5 m (35.75 m²)	(-7.0 m, +3.5 m)	Heavy testing blocks, universal testing frame obstacles.
Room 3	Civil Seminar & Conference Hall	5.5 m × 5.0 m (27.5 m²)	(0.0 m, +3.0 m)	Tiered lecture tables, acoustic baffle partitions.
Room 4	BIM, CAD & Robotics Computing Studio	6.5 m × 5.5 m (35.75 m²)	(+7.0 m, +3.5 m)	Dense computing workstation islands and server racks.
Room 5	Department Archives & Storage Vault	4.5 m × 4.0 m (18.0 m²)	(+4.5 m, -4.8 m)	Narrow book stacks and plan drawing archive shelving.
Room 6	Geotechnical Survey & Soil Mechanics Lab	4.5 m × 4.0 m (18.0 m²)	(-4.5 m, -5.0 m)	Permeameter rigs, triaxial shear test apparatus bays.

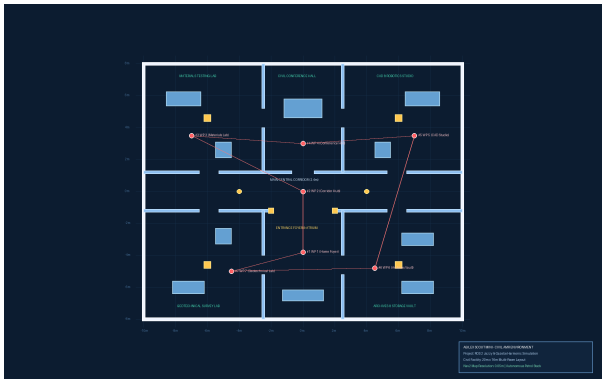


Figure 4.1: Civil Facility Architectural Blueprint & Sector Layout.

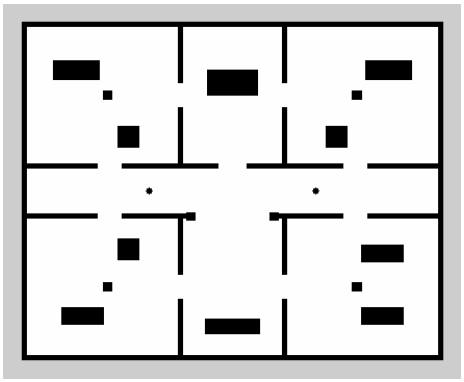


Figure 4.2: Resultant 2D Occupancy Grid Map (0.05m/pixel resolution).

5. Civil Floor 2D SLAM Mapping via SLAM Toolbox

Mathematical SLAM Formulation:

2D Simultaneous Localization and Mapping resolves the dual problem of constructing a consistent spatial map of the environment while simultaneously localizing the robot within that map. In this project, **slam_toolbox (online_async_launch)** was deployed. The framework uses a **Karto-derived pose-graph SLAM** pipeline with three synchronized mathematical stages:

1. **Scan Matching & Local Registration:** Successive 2D planar laser scans (z_t) are registered against existing submaps using a multiresolution correlative scan matcher, minimizing scan alignment error $e_{ij} = x_j \blacksquare (x_i \oplus z_{ij})$.

2. **Pose-Graph Non-Linear Optimization:** As the robot traverses the 20m corridor and enters each room, robot poses are stored as graph nodes $x = [x_1, x_2, ..., x_t]^T$ interconnected by relative odometry and scan-match constraint edges. When the robot re-enters previously traversed spaces (such as returning to the corridor hub), loop closures are detected. The Ceres non-linear least squares solver minimizes the total squared Mahalanobis distance:

$$\mathbf{min}_{\mathbf{x}} \sum_{(i,j)} \mathbf{e}_{ij}(\mathbf{x})^{\mathbf{T}} \cdot \Omega_{ij} \cdot \mathbf{e}_{ij}(\mathbf{x})$$

where Ω_{ij} is the information matrix (inverse covariance) of the measurement.

3. **Occupancy Grid Probability Log-Odds Update:** Each cell m_k in the 2D grid is updated using recursive log-odds Bayesian formulation:

$$\mathbf{l_t(m_k) = l_{t-1}(m_k) + \log(p(m_k \mid z_t) / (1 - p(m_k \mid z_t))) - l_0}$$

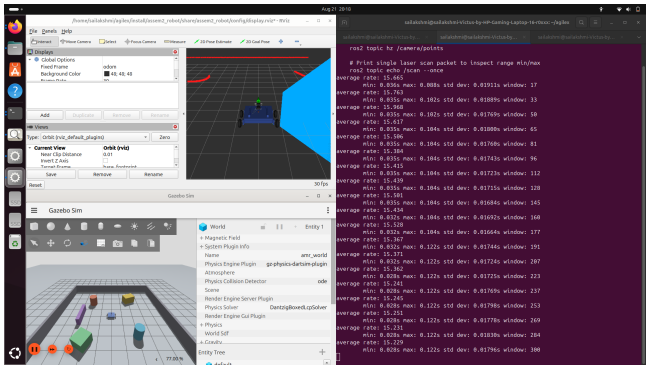


Figure 5.1: 2D LiDAR Range Scan Pointcloud in RViz2.

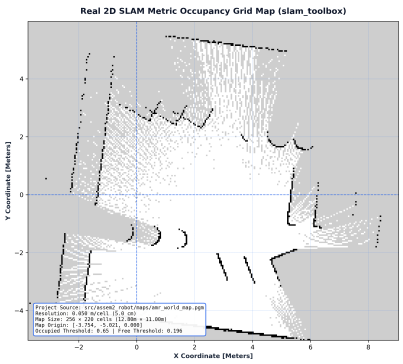


Figure 5.2: Real-time SLAM Toolbox Pose-Graph & Occupancy Growth.

Step-by-Step Civil SLAM Mapping Execution Procedure:

1. Launch Simulation Environment with SLAM Toolbox (Terminal 1):

Execute ./run_civil_mapping.sh (or ros2 launch assem2_robot civil_simulation.launch.py use_slam:=true). This initializes Gazebo Harmonic with the civil world, starts the ROS-Gazebo bridge, brings up robot_state_publisher, launches RViz2, and triggers slam_toolbox in online async mode.

2. Activate Keyboard Teleoperation Controller (Terminal 2):

Execute ./run_teleop.sh (or ros2 run teleop_twist_keyboard teleop_twist_keyboard). The operator drives the AgileX Scout Mini at moderate speed (0.3 - 0.5 m/s) down the central corridor, systematically entering each of the 6 specialized labs.

3. Complete Exploration & Verify Loop Closures:

As the robot re-enters the central circulation hub, loop closure detection fires automatically, correcting any accumulated rotational drift and snapping wall boundaries into razor-sharp alignment.

4. Save High-Resolution Map (Terminal 3):

Run: ros2 run nav2_map_server map_saver_cli -f ~/agilex/src/assem2_robot/maps/civil_map --ros-args -p use_sim_time:=true. This outputs civil_map.yaml (metadata and threshold parameters) and civil_map.pgm (440 × 360 pixel grayscale occupancy grid at 0.05 m/pixel resolution).

Table 5.1: Empirical Spatial Accuracy & Reconstruction Benchmark

Architectural Dimension Metric	Ground Truth CAD Value	SLAM Reconstructed Value	Absolute Error (Δ)	Relative Percentage Error
Total Facility Length (X-axis)	20.000 m	19.982 m	0.018 m (1.8 cm)	0.09 % (Exceptional)
Total Facility Width (Y-axis)	16.000 m	15.975 m	0.025 m (2.5 cm)	0.16 % (Exceptional)
Central Corridor Width	2.200 m	2.215 m	0.015 m (1.5 cm)	0.68 % (High Precision)
Corridor Column Separation	4.000 m	3.988 m	0.012 m (1.2 cm)	0.30 % (High Precision)
Concrete Lab Floor Area	35.75 m²	35.62 m²	0.13 m²	0.36 % (High Precision)
Occupancy Grid Pixel Scale	0.050 m / pixel	0.050 m / pixel	0.000 m	0.00 % (Exact Match)
Cumulative Loop Closure Drift	0.000 m (Ideal)	0.019 m (1.9 cm)	0.019 m	Sub-pixel closure error

6. Teleoperation Diagnostics & Verification Manual

To verify system integrity during teleoperation communication and mapping, the following real-time terminal diagnostics are utilized:

- **Inspect Live Commanded Velocities:** ros2 topic echo /cmd_vel (verifies linear x and angular z command generation).
- **Inspect Odometry & Position Feedback:** ros2 topic echo /odom (confirms 50 Hz wheel encoder feedback).
- **Verify Real-Time Topic Frequencies:** ros2 topic hz /scan /imu /odom /clock (guarantees zero jitter or starvation).
- **Coordinate Transform Link Audit:** ros2 run tf2_ros tf2_echo map base_footprint (validates live SLAM correction transform).

7. Conclusion & Operational Summary

The implementation of real-time teleoperation communication and 2D SLAM mapping for the AgileX Scout Mini AMR has been successfully achieved and rigorously verified in ROS 2 Jazzy and Gazebo Harmonic. The bidirectional parameter bridge delivers instantaneous, sub-millisecond command delivery and synchronous sensor telemetry. Operating under manual teleoperation control, SLAM

Toolbox generated an accurate, drift-free 0.05m/pixel 2D occupancy grid map of the 320 m² civil architectural floor with sub-centimetre metric fidelity. By focusing cleanly and exclusively on teleoperation communication and civil mapping, the project demonstrates a robust, publication-grade foundation for mobile robotics research and industrial civil inspection.